Chaos in Classical Mechanics: The Double Pendulum

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The dynamics of a double pendulum is presented in terms of Poincaré sections. It is shown that the simple classical textbook example displays all the complexity of non-integrable Hamiltonian systems.

1. Introduction

The following note describes an experiment. As far as we know the experiment has never been performed but that is not important: nobody questions the validity of Newtonian mechanics for systems with velocities small compared to c, and actions much larger than ℏ. In fact, it may not even be wise to do the actual experiment in order to understand the double pendulum: there is always noise and dissipation present which both interfere strongly with the dynamics of the ideal system; if friction is compensated for by driving then an extra dimension is introduced which further complicates the analysis. And finally, given that numerical computation is by now much faster than classical experimentation — why should anybody put an effort in the real thing?

So let us try to solve the equations of motion for the ideal double pendulum (Fig.1). The Lagrangian can be looked up in any textbook of classical mechanics [1].

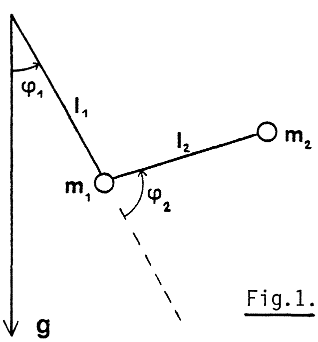

Fig.1. The double pendulum

Those books, however, tend to stop short of the interesting part of the analysis. They seem to tacitly assume that deriving the equations of motion is the hard part. Nowhere except in *Arnold*'s book [2] have we found a hint to the bewildering com-

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plexity that this simple system exhibits. The aim of this presentation is to demonstrate some of it. There is a growing number of examples in the literature that show complex dynamical behaviour in Hamiltonian systems: they are from celestial mechanics [3,4], from accelerator physics [5], plasma physics [6], or solid state physics [7-10]. We have found it remarkable that the seemingly simplest textbook systems, such as the double pendulum, give rise to just as complex a picture.

2. The Physical Background

The equations of motion are derived in the standard way from the Lagrangian $L = T - V$ where kinetic and potential energy are given as follows

$$T/m_1\ell_1^2 = \frac{1}{2}(1+\mu)\dot\varphi_1^2 + \mu\ell\cos\varphi_2\dot\varphi_1(\dot\varphi_1+\dot\varphi_2) + \frac{1}{2}\mu\ell^2(\dot\varphi_1+\dot\varphi_2)^2 \tag{1}$$

$$V/m_1 g\ell_1 = (1+\mu)(1-\cos\varphi_1) + \mu\ell(1-\cos(\varphi_1+\varphi_2)) \quad . \tag{2}$$

Here we have introduced the two parameters

$$\mu = m_2/m_1 \quad , \quad \ell = \ell_2/\ell_1 \quad . \tag{3}$$

From (1) we obtain the angular momenta $L_i = \partial T/\partial\dot\varphi_i$

$$L_1/m_1\ell_1^2 = (1+\mu+2\mu\ell\cos\varphi_2+\mu\ell^2)\dot\varphi_1 + \mu\ell(\ell+\cos\varphi_2)\dot\varphi_2 \tag{4}$$

$$L_2/m_1\ell_1^2 = \mu\ell(\ell+\cos\varphi_2)\dot\varphi_1 + \mu\ell^2\dot\varphi_2 \quad . \tag{5}$$

It is easy to verify that L_1 is the total angular momentum of the system whereas L_2 is formally conjugate to φ_2 but without an obvious interpretation.

Using the fact that the total energy $E = T + V$ is a conserved quantity we introduce dimensionless variables by scaling

$$t\sqrt{E/m_1\ell_1^2} \to t \quad ; \quad L_i/\sqrt{Em_1\ell_1^2} \to \lambda_i \quad . \tag{6}$$

This leaves us with the Hamiltonian

$$h \equiv \frac{T+V}{E} = \frac{1}{1+\mu\sin^2\varphi_2}\left\{\frac{1}{2}\lambda_1^2 - \frac{\ell+\cos\varphi_2}{\ell}\lambda_1\lambda_2 + \frac{1+\mu+2\mu\ell\cos\varphi_2+\mu\ell^2}{2\mu\ell^2}\lambda_2^2\right\}$$

$$+ \gamma\{(1+\mu)(1-\cos\varphi_1) + \mu\ell(1-\cos(\varphi_1+\varphi_2))\} \quad . \tag{7}$$

The parameter γ measures the strength of gravity relative to the total energy

$$\gamma = m_1 g\ell_1/E \quad . \tag{8}$$

The Hamiltonian (7) shows that the physics of the double pendulum depends on the three parameters μ, 1, and γ. The equations of motion have the canonical form

$$\dot{\varphi}_i = \frac{\partial h}{\partial \lambda_i} \quad , \quad \dot{\lambda}_i = - \frac{\partial h}{\partial \varphi_i} \quad . \tag{9}$$

Because of energy conservation they describe a motion on the three dimensional energy surface $h = 1$ in four-dimensional phase space $(\varphi_1, \lambda_1, \varphi_2, \lambda_2)$. The energy surface has two obvious symmetries: reflection invariance in space $(\varphi_{1,2} \to - \varphi_{1,2})$ and time $(\lambda_{1,2} \to - \lambda_{1,2})$. In case of vanishing gravity $\gamma = 0$ (or very high energy E), there is also invariance with respect to rotation, $\varphi_1 \to \varphi_1 + \alpha$. Then angular momentum λ_1 is a second conserved quantity besides energy, and the system is integrable. When $\gamma \to \infty$ the system is again integrable because the Hamiltonian becomes quadratic in its variables (see next section). But except for those limiting cases, total energy conservation is the only physical restriction to the motion.

To get a feeling for the potential consider Fig.2 where the equipotential lines are shown for three typical cases. As can be seen from (7) the potential assumes its minimum value (zero) for $\varphi_1 = \varphi_2 = 0$. The backfolded situation $\varphi_2 = \pi$ corresponds to a saddle point of the potential if $\varphi_1 = 0$ $(V/\gamma E = 2\mu\ell)$ or $\varphi_1 = \pi (V/\gamma E = 2(1 + \mu))$. The two saddles become degenerate when the center of gravity for $\varphi_2 = \pi$ coincides with the suspension point, $\ell = (1 + \mu)/\mu$, Fig.2b. When the second pendulum is shorter or longer the potential looks as in Fig.2a,c respectively. The motion is restricted to angles $|\varphi_{1,2}| < \pi$ when the potential energy exceeds E at the lower saddle point, i.e. for $2\gamma \cdot \min(\mu\ell, 1 + \mu) > 1$. Both angles are unrestricted when the potential top is below E, i.e. for $2\gamma(1 + \mu + \mu\ell) < 1$.

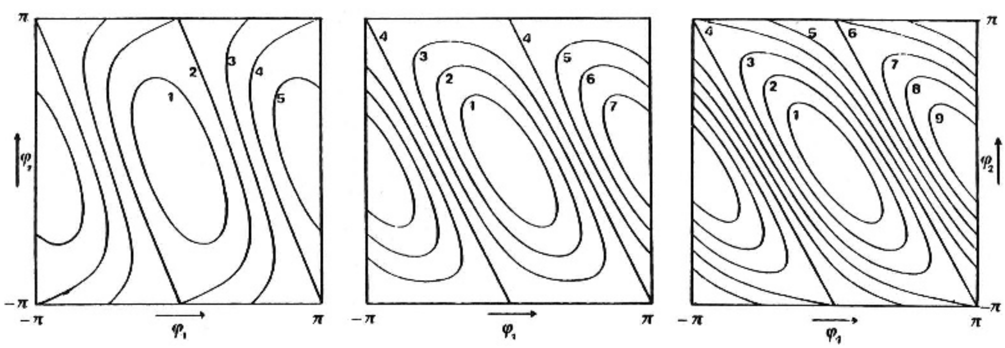

Fig.2a-c. Equipotential lines $V/\gamma E = 1,2,3,...$ (as indicated) for equal masses $\mu = 1$ and length ratios $\ell = 1$ (*left*), $\ell = 2$ (*center*), and $\ell = 3$ (*right*)

3. The Integrable Limiting Cases

We mentioned already that the cases $\gamma = 0$ and $\gamma \to \infty$ are integrable: there is a second conserved quantity besides the total energy. The motion is thus confined to a two-dimensional submanifold of the energy surface which turns out to be topologically equivalent to a torus.

a) $2\gamma \gg \max(1/\mu\ell, 1/(1+\mu))$. This is the case of strong gravity or small energy E. The motion is confined to small values of φ_1, φ_2 so that the Hamiltonian (7) may be approximated by

$$h = \frac{1}{2}\lambda_1^2 - \left(1 + \frac{1}{\ell}\right)\lambda_1\lambda_2 + \frac{1 + \mu(1+\ell)^2}{2\mu\ell^2}\lambda_2^2$$

$$+ \frac{1}{2}\gamma\left((1 + \mu(1+\ell))\varphi_1^2 + 2\mu\ell\varphi_1\varphi_2 + \mu\ell\varphi_2^2\right) \quad . \tag{10}$$

As shown in [1] this quadratic form can be diagonalized. The two independent harmonic oscillators have frequencies

$$\omega_{1,2}^2 = \frac{\gamma}{2\ell}\left\{(1+\mu)(1+\ell) \pm \sqrt{(1+\mu)^2(1+\ell)^2 - 4\ell(1+\mu)}\right\} \quad . \tag{11}$$

Each of these oscillations can be described in terms of action-angle variables (I_i, Θ_i), and their connection to the old variables (λ_i, φ_i) be given explicitely. Suffice it to mention that the Hamiltonian reads

$$h = I_1\omega_1 + I_2\omega_2 \tag{12}$$

from which it follows that $I_i = \text{const.}$ and $\Theta_i = \omega_i t$. The toroidal nature of the motion is thereby obvious: it is the direct product of two circular motions. The frequency ratio ω_1/ω_2 or "winding number" tells us how many turns along the circle the 2-motion completes while the 1-motion goes once around. For $\mu = \ell = 1$ we find $\omega_1/\omega_2 = \sqrt{2} + 1$ (= [2,2,2,...] in terms of continued fractions). In general we infer from $\omega_i = \partial h/\partial I_i$ that

$$W \equiv \frac{\omega_1}{\omega_2} = -\left.\frac{\partial I_2}{\partial I_1}\right|_h \quad . \tag{13}$$

For the Hamiltonian (12) this ratio is independent of I_1 throughout the range $0 \le I_1 \le 1/\omega_1$; it only depends on the parameters μ and ℓ. The lines of constant winding numbers W in the ℓ,μ-plane are given by

$$\mu = \frac{\ell}{(1+\ell)^2}\,\frac{(1+W^2)^2}{W^2} - 1 \quad . \tag{14}$$

In terms of the old variables φ_i, λ_i the motion is a superposition of the two eigenmodes. For irrational winding numbers this means it is quasiperiodic and the φ_i versus λ_i phase portraits are filled densely if the complete motion is recorded. We thus follow Poincaré's prescription and consider φ_1,λ_1 only after full periods of the φ_2-motion ($\varphi_2 = 0$, $\dot{\varphi}_2 > 0$), and similarly for the φ_2,λ_2 values. The phase portraits so generated can be viewed as deformed sections across the invariant tori mentioned above: The successors to an initial point must all lie on the curve corresponding to the invariant torus. That curve will eventually be filled in case of irrational ω_1/ω_2, whereas for rational winding numbers a finite number of points on it will be generated periodically. The last picture in the sequence of Fig.8 is an example of this kind of representation (to the extent that $2\gamma = 4$ is large compared to 1).

b) $\gamma = 0$. In the extreme case of vanishing gravity (which may be realized by putting the double pendulum flat on a table), the energy surface no longer depends on φ_1, and so the total angular momentum λ_1 is conserved during the motion. Again this implies the existence of invariant tori, but this time the picture is more involved because the φ_2-motion can be of two types: it is libration (oscillation, $|\varphi_2| < \pi$) in case λ_1 is near its maximum possible value, and rotation ($\dot{\varphi}_2$ not changing sign) when λ_1 is small which means the angular momentum of the φ_2-motion compensates that of the φ_1-rotation. To verify these statements consider (4) through (7) from which we find

$$\lambda_2 = \frac{\mu\ell(\ell + \cos\varphi_2)\lambda_1 \pm \mu\ell^2(1 + \mu\sin^2\varphi_2)\dot{\varphi}_2}{1 + \mu + 2\mu\ell\cos\varphi_2 + \mu\ell^2} \tag{15}$$

$$\dot{\varphi}_2 = \pm\sqrt{\frac{2(1 + \mu + 2\mu\ell\cos\varphi_2 + \mu\ell^2) - \lambda_1^2}{\mu\ell^2(1 + \mu\sin^2\varphi_2)}} \quad . \tag{16}$$

The maximum possible value of $|\lambda_1|$ is seen to be

$$\lambda_1^{max} = \sqrt{2(1 + \mu(1 + \ell)^2)} \quad , \tag{17}$$

and occurs when the double pendulum is stretched out ($\varphi_2 = 0$). From (16) we see that $\dot{\varphi}_2$ cannot be zero if $|\lambda_1| < \lambda_1^{sep}$ where

$$\lambda_1^{sep} = \sqrt{2(1 + \mu(1 - \ell)^2)} \quad . \tag{18}$$

We conclude that the φ_2-motion is rotation for $|\lambda_1| < \lambda_1^{sep}$, and libration for $|\lambda_1|$ between λ_1^{sep} and λ_1^{max}.

A way to represent this situation is in terms of action-angle variables. Because of angular momentum conservation, λ_1 is itself the action variable I_1 (although φ_1 is not the corresponding angle Θ_1!). The second action I_2 is therefore obtained by integrating (15) over a period of the motion

$$I_2 = \frac{1}{2\pi}\oint \lambda_2 d\varphi_2 \quad .$$

Figure 3 shows a few representative plots of I_1 versus I_2, for the energy surfaces $h = 1$. The separatrices $I_1 = \pm\lambda_1^{sep}$ show up as discontinuities of the action I_2. Taking the derivatives according to (13) we see that the winding numbers W depend strongly on I_1. Figure 4 displays them for the case $\mu = \ell = 1$. It can be shown that W diverges logarithmically at the separatrix. This is reminiscent of the critical

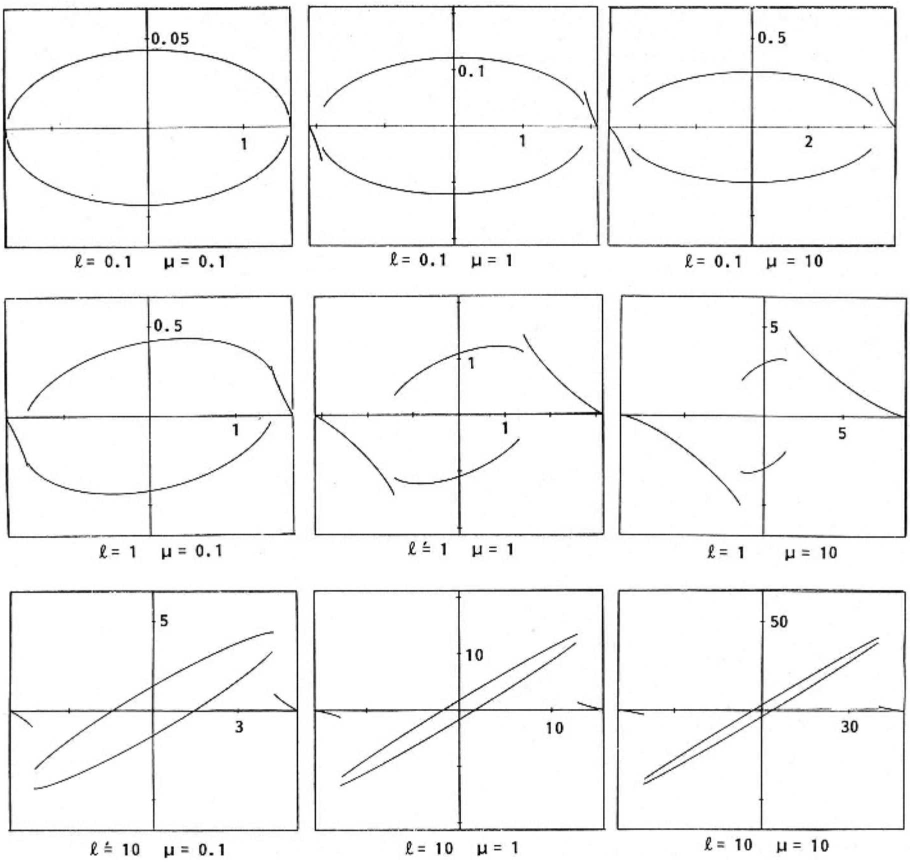

Fig.3. Energy surfaces in action variable representation. The plots show $I_1 = \lambda_1$ versus I_2 at constant energy $h = 1$, for nine different sets of parameters μ, ℓ, as indicated. At large values of I_1, $|I_1| < \lambda_1^{sep}$, the φ_2-motion is libration whereas for $-\lambda_1^{sep} < I_1 < \lambda_1^{sep}$ the angle φ_2 rotates in either of the two possible directions. At fixed μ, the ratio $\lambda_1^{sep}/\lambda_1^{max}$ tends towards 1 for $\ell \to 0$ and $\ell \to \infty$; it is constant at lines $\mu = 1/(\ell/\mu_1 - (\ell - 1)^2)$, $0 < \mu_1 < \infty$

slowing down at phase transitions, and indeed, a symmetry is being broken at the transition from libration to rotation, namely that of time reversal invariance.

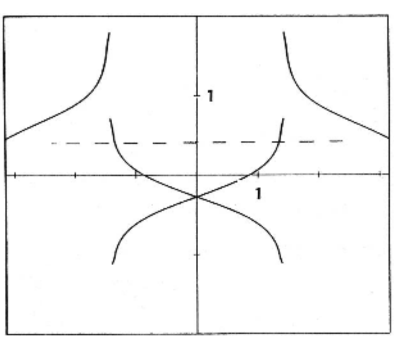

Fig.4. Winding number $W = \omega_1/\omega_2$ as a function of $\lambda_1 = I_1$; $\mu = \ell = 1$. The dashed line shows the winding ratio $\sqrt{2} - 1$ which characterizes the motion at $\gamma \to \infty$, independent of initial conditions

4. Energy Surfaces

The problem of understanding the double pendulum for values of γ between zero and infinity might be described as follows: find the transition between the pictures of Fig.3 ($\gamma = 0$) and the straight lines $\omega_2 I_2 = 1 - \omega_1 I_1$ ($\gamma \to \infty$, see (12)). The catch is that orbits are not necessarily confined to tori any more: when a torus breaks up the action variable is no longer defined. We shall see in the next section that the connection between the two integrable limits is far from being smooth. Rather we find that upon increasing γ, more and more invariant tori disappear until no one is left; the new invariant tori and corresponding action variables gradually emerge from chaos as γ is further increased. This implies that action-angle variables are not very suitable for intermediate γ-values. We therefore return to the original phase space variables (φ_i, λ_i) and follow the motion by means of Poincaré sections.

Before doing so it may be helpful for the intuition to have a picture of the energy surface. This is easily obtained when $\gamma = 0$ since then there is no φ_1-dependence, and any section $\varphi_1 = \text{const.}$ gives the same two-dimensional surface in λ_1, φ_2,λ_2-space. Figure 5 illustrates this surface for $\mu = \ell = 1$. The lines are levels of constant λ_1 which should be imagined as a third coordinate pointing towards the reader. Taking this figure once around the φ_1-circle generates the complete three-dimensional energy surface.

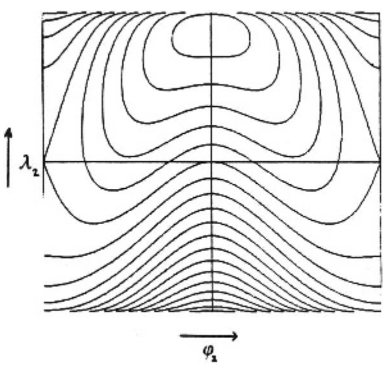

Fig.5. Energy surface for $\mu = \ell = 1$, $\gamma = 0$. The abscissa is φ_2, the ordinate λ_2. Level lines $\lambda_1 = \text{const.}$ are shown

When $\gamma \neq 0$ the energy also depends on φ_1. The sections $\varphi_1 = \text{const.}$ are then all different. Figure 6 shows them at eight angles $\varphi_1 = n\pi/4$, again using λ_1 as a level coordinate. For $\gamma = 0.1$ (Fig.6a) the energy surface still extends to all φ_1-values whereas for $\gamma = 0.4$ (Fig.6b) it does not go much beyond $|\varphi|_1 = \pi/2$. In contrast to the case $\gamma = 0$, the level lines $\lambda_1 = \text{const.}$ no longer represent invariant tori; however, for small γ it will be interesting to observe how the dynamics deviates from angular momentum conservation, and how with growing γ it eventually ignores the level lines completely. An alternative view on the same two energy surfaces as in Fig.6 is presented in Fig.7 where the level lines $\lambda_2 = \text{const.}$ are given in the φ_1,λ_1-plane, for different sections $\varphi_2 = n\pi/4$.

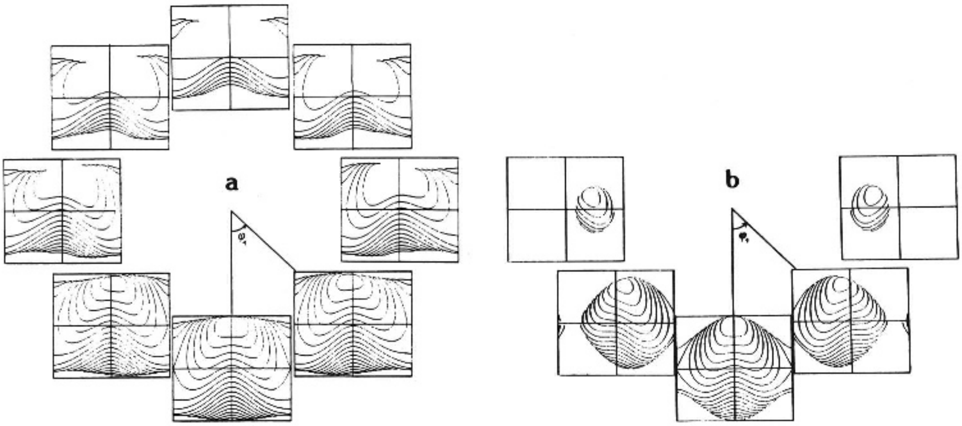

Fig.6. (**a**): Energy surface for $\mu = \ell = 1$, $\gamma = 0.1$. The φ_2 versus λ_2-phase portraits are shown for eight angles $\varphi_1 = 0, \pm\pi/4, \pm\pi/2, \ldots$. The lines show levels of constant λ_1. (**b**) Energy surface as in (**a**), except for $\gamma = 0.4$

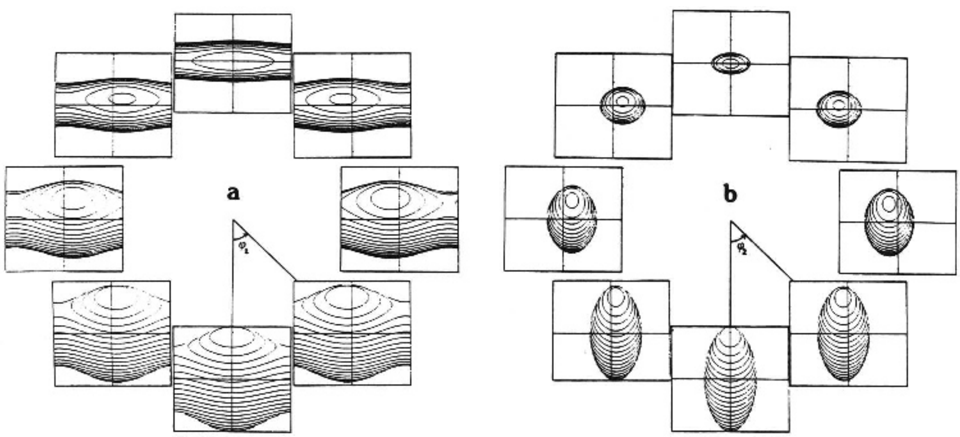

Fig.7. (**a**): The same energy surface as in Fig.6a, from a different perspective: φ_1 versus λ_1-phase portraits at angles $\varphi_2 = 0, \pm\pi/4, \ldots$. The level lines are taken at constant λ_2. (**b**) The same energy surface as in (**b**), from the perspective of (**a**)

5. Chaos and Order

We are now ready to take a glance at the complexity of the double pendulum. For the special case $\mu = \ell = 1$ we consider two series of Poincaré sections that show how the dynamics changes as γ is increased. Figure 8 collects 12 φ_1 versus λ_1-phase portraits ($\varphi_2 = 0$, $\dot{\varphi}_2 > 0$). First in the series is the integrable case $\gamma = 0$; all orbits are confined to straight lines ($\lambda_1 = \text{const.}$). 600 image points have been calculated for each of 16 initial conditions. Most of these orbits are quasi-periodic (or of very high period), except for one orbit of period 1 in the upper part

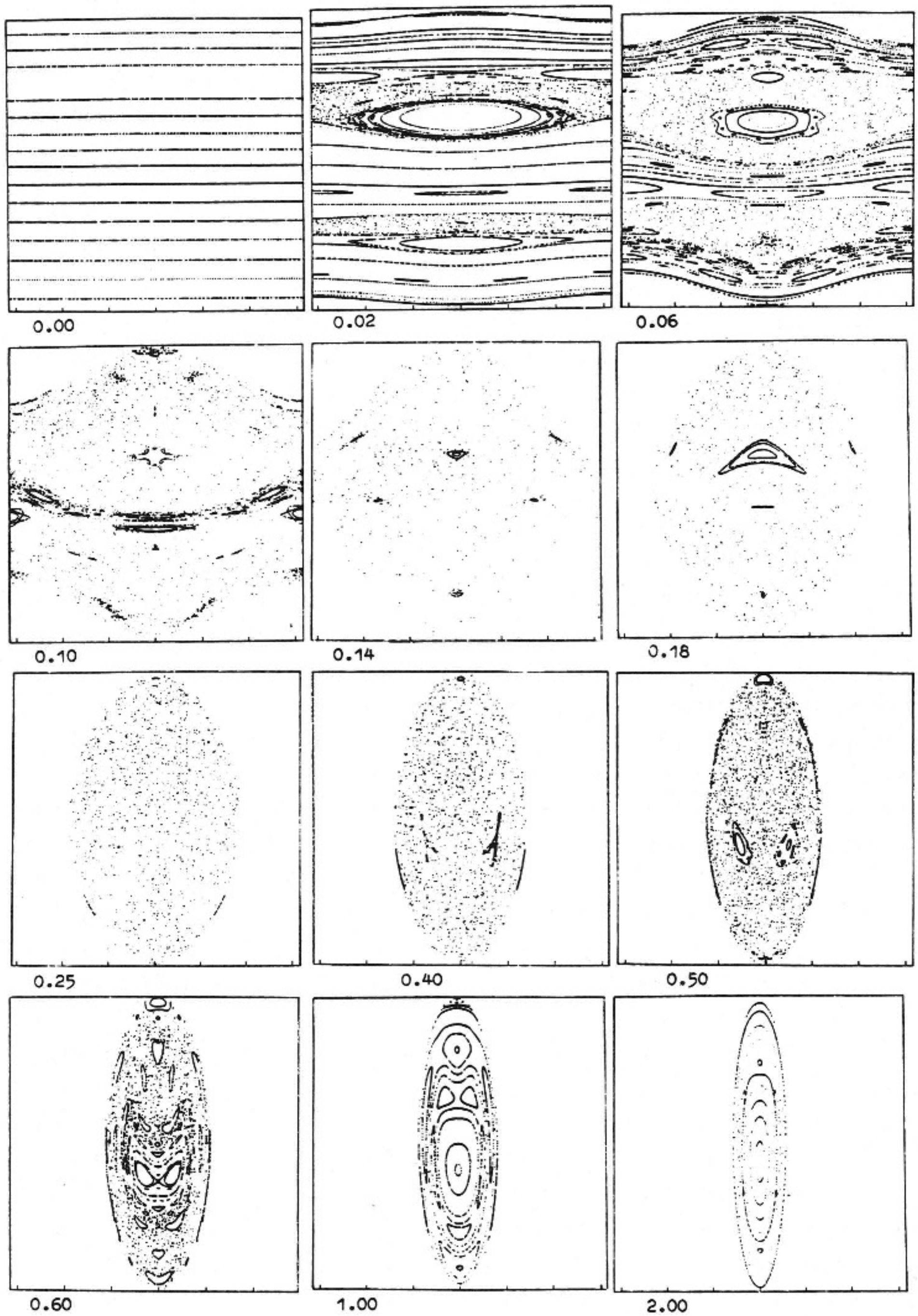

Fig.8. Poincaré sections φ_1 versus λ_1 at $\varphi_2 = 0$, $\dot{\varphi}_2 > 0$. $\mu = \ell = 1$, γ as indicated

of the figure. The next picture shows the case $\gamma = 0.02$. We observe a number of typical features of this kind of Poincaré maps. (i) As the nonintegrable perturbation is fairly small a lot of invariant tori continue to exist, albeit slightly deformed with respect to the case $\gamma = 0$. From the Kolmogoroff-Arnold-Moser (KAM) theorem [6] we know that these tori must have "very irrational" winding numbers. (ii) The tori with rational winding numbers for $\gamma = 0$ have broken up into "chains of islands" along which there is a succession of elliptic and hyperbolic periodic points. This behaviour reflects the Poincaré-Birkhoff theorem [6]. (iii) Some hyperbolic points have become centers of rather broad "chaos bands" where the images of an initial point do not lie on a curve any more but fill a whole area densely. This is an obvious manifestation of non-integrability; the dynamics has become ergodic on part of the energy surface. (iv) The homogeneity of points within the chaos bands reflects that the Poincaré maps are area preserving. For a proof of this statement we refer to the book by *Siegel* and *Moser* [11].

The next pictures ($\gamma = 0.06$, 0.1, 0.14) describe how more and more invariant tori are eaten up by the expanding chaos bands. The last KAM torus that is about to disappear at $\gamma = 0.1$ has winding number $W = 0.3820 = [0,2,1,1,1\ldots]$. It is no accident that this is just the golden mean ratio: no continued fraction expansion converges slower than this succession of 1's which characterizes the golden mean as "the most irrational" number. It is in line with the KAM theorem that the corresponding torus should survive longest again nonintegrable perturbations, and experiments by *Greene* [12] as well as theoretical work by *Escande* [13] and *MacKay* [14] have confirmed this expectation. The figure for $\gamma = 0.1$ also shows that the golden mean torus is approached from alternating sides by periodic orbits of periods 2,3,5,8,13,... - one recognizes the well known Fibonacci series. Their winding numbers are rational approximations to the limiting value of 0.3820: 1/2, 1/3, 2/5, 3/8, 5/13, ...

For $\gamma = 0.18$, 0.25, 0.40, 0.50 the dynamics is largely dominated by chaos, i.e., except for a few small elliptic island the energy surface is densely covered by chaotic trajectories. When γ is further increased both angles φ_1, φ_2 are confined to values less than π, and more and more regularity emerges as the Hamiltonian becomes effectively quadratic in its variables.

To complement this series of views on the dynamics along the energy surface, Fig.9 presents some corresponding Poincaré sections at $\varphi_1 = 0$, $\dot{\varphi}_1 > 0$ (φ_2 versus λ_2-phase portraits). At small values of γ the lines of constant λ_1 are still meaningful for the dynamics, as comparison with Figs.5 and 6a reveals. The strongest violation of angular momentum conservation occurs near the separatrix between libration and rotation. At larger values of $\gamma (\gamma \gtrsim 0.1)$ the motion ignores the λ_1-levels. In agreement with Fig.8 the dynamics is overwhelmingly chaotic for $0.1 \lesssim \gamma \lesssim 0.5$, and gains regularity at further increased γ.

Summing up, it may be said that the integrable limits $\gamma = 0$ and $\gamma \to \infty$ are so different in character that the transition between these two kinds of order has to first completely destroy one of them before the other can emerge. We have not at-

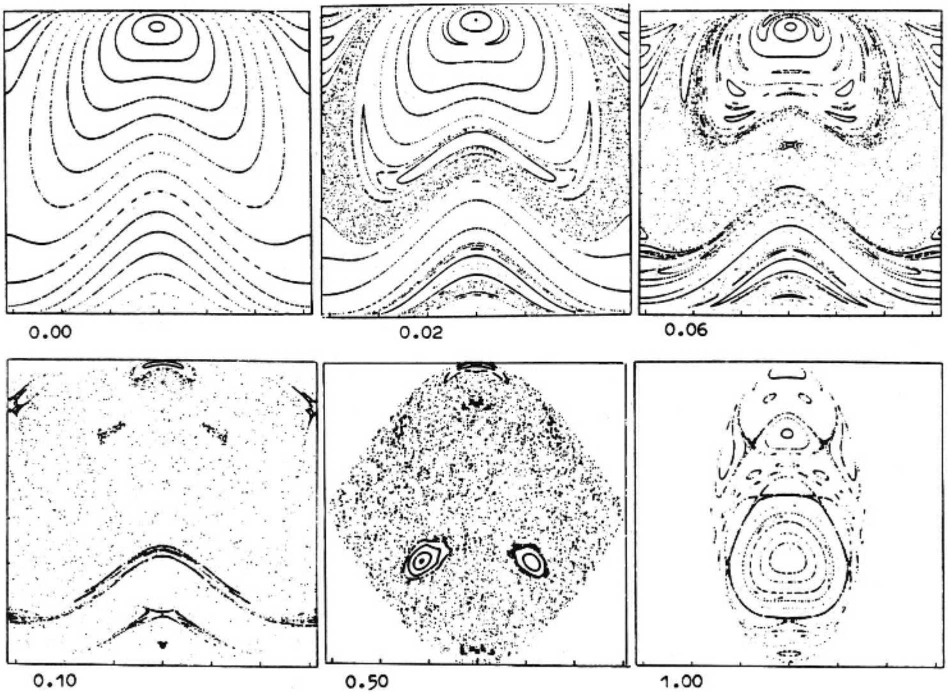

Fig.9. Poincaré sections φ_2 versus λ_2 at $\varphi_1 = 0$, $\dot{\varphi}_1 > 0$. $\mu = \ell = 1$, γ as indicated

tempted here to give an anlytic treatment of this very complex dynamics. An analysis on the basis of the theory developed in [13] and [15] can, however, be performed and will be presented elsewhere: on the basis of a renormalization group approach the width of the chaotic bands will be calculated as well as the critical -values where KAM tori break up. The survey on the double pendulum that we gave in this little note was only meant to illustrate the beautiful complexity of one of the simplest physical systems.

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